SMARAK SAMAL

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Education

Birla Institute of Technology And Science (BITS) Pilani, Hyderabad Campus — Hyderabad
Bachelor of Engineering in Mechanical, Master of Science in Chemistry — *Oct. 2022 – June 2027*

Experience

Research Intern—Machine Learning — Aug. 2026 – Dec. 2026
BITS Pilani — *Hyderabad*

- Developing Deep Learning models **(CNN/Transformer)** for real-time Anomaly Detection in Vibration Signal Data,
- Building Robust Pipelines for time-series signal Pre-processing, feature extraction, classification & Designing Abort prediction systems using sensor data to enable proactive fault detection in industrial/mechanical systems.

Data Analyst Intern — May 2024 – July 2024
Indian Red Cross Society — *Kolkata*

- Built an automated weekly reporting pipeline (Python + Excel) replacing manual data consolidation, **saving 12 hours/week** of staff time across 3 program teams
- Analyzed donor retention patterns using SQL and Python across a dataset of **50,000+ donor records**, identifying a **23% drop-off** after first donation and enabling a targeted re-engagement campaign that improved repeat donation rate **by 18%**

Projects

Product Growth & User Retention System of Netflix | *Product Analytics*

- Analyzed **100K+ user interactions** from a Netflix-style streaming platform using SQL and Python to evaluate acquisition, activation, retention, and engagement metrics.
- Developed cohort retention models and churn segmentation frameworks, uncovering user behavior patterns across **12 monthly cohorts** and reducing retention analysis **time by 60%.**

Market Research & Analysis on Hinge | *Product Management*

- Identified a **65% chat drop-off after 2 messages** via funnel analysis and validated root causes through **30+ user surveys and chat replays** — translating data insights into prioritised product decisions
- Designed a native Date Planner feature with **4+ measurable success metrics** (chat-to-date conversion, retention, monetisation); proposed A/B test framework that projected a **20% lift in chat-to-date conversion rate**

Content-Based Movie Recommendation System | *Machine Learning*

- Built a content-based recommender over **4,800+ TMDB movies**, combining genre, keyword, cast, crew and plot features into text vectors (CountVectorizer, 5,000 features) and ranking titles by cosine similarity.
- Ranked titles by cosine similarity on 5,000-feature bag-of-words vectors (CountVectorizer, English stop words removed) over a **4,806 × 4,806** similarity matrix, returning the top 5 similar movies per query.

Corporate Sales Pipeline & Revenue Forecasting Platform | *CRM Performance Optimization*

- Built lead-scoring model predicting sales conversion with **84% accuracy**.Automated CRM performance monitoring for **50K+ sales opportunities**.
- Reduced lead response delays by identifying high-priority prospects.Created KPI dashboard tracking pipeline health and sales productivity.

Position of Responsibility

Sponsorship Lead — Fashion Club, BITS Pilani, Hyderabad — May 2024 - May 2025

- Secured 15+ lakh in sponsorships by engaging 100+ companies, achieving a high conversion rate through strategic pitching and negotiations.
- Led a team of **40+ members**, managed end-to-end sponsor relations, ensuring seamless execution and **100% fulfillment** of contractual deliverables.

Vice President — Utkal Samaj, BITS Pilani, Hyderabad — May 2024 - May 2025

- Managed planning & execution of 5+ cultural events for 3,000+ attendees, leading a 40+ member volunteer & core team. Led budget tracking, vendor relations & logistics for on-time, cost-controlled execution across 5+ cultural events

Social Media and Outreach Lead — Hindi Tarang, BITS Pilani, Hyderabad — May 2024 - May 2025

- Led data-driven digital outreach; streamlined strategy to **boost reach & engagement** by **20%** via **3-4 weekly posts**

Technical Skills

Languages & Frameworks : C++, Java, Python (Pandas, NumPy, Matplotlib,scikit-learn,statmodels), SQL
Analytics & BI : A/B Testing, Funnel & Cohort Analysis,Hypothesis testing, churn and LTV modelling, Power BI, Tableau, Excel
ML : recommendation systems, ranking, classification, time-series modelling, LLM API integration,segmentation,forecasting
AI / LLM : LLM API integration (OpenAI, Claude), Prompt Engineering, Agentic AI